

OIL INDIA LIMITED — DRILLING DIRECTORATE

DULIAJAN, ASSAM · WELL COMPLETION REPORT (WCR) · SECTION 8

Well Name:	MORAN-7 (MOR-07)	Field / Block:	Moran PML (MOR-I)
Spud Date:	10-MAR-2018	TD Reached:	22-AUG-2018 (3,450.0m MD)
Rig Assigned:	OIL-RIG-01 (2000 HP AC-SCR)	Status:	COMPLETED PRODUCER
Target Horizon:	Barail Group / Tipam Sandstone	Classification:	RESTRICTED SUBSURFACE ASSET

SECTION 4: GEOLOGICAL STRATIGRAPHY & FORMATION TOPS

Formation Name	Depth Top (m MD)	Depth Base (m MD)	Lithology & Characteristics
Alluvium / Dihing	0.0 m	290.0 m	Unconsolidated sand, gravel, surface water table
Dhekiajuli Formation	290.0 m	790.0 m	Coarse friable sands, pebble beds, seepage zone
Namsang Formation	790.0 m	1,490.0 m	Sandstones with interbedded soft claystone layers
Girujan Clay Formation	1,490.0 m	2,180.0 m	Thick sticky swelling montmorillonite clay. High sticking risk.
Tipam Sandstone (TS-1 to TS-6)	2,180.0 m	2,960.0 m	Multi-storied braided river pay sands. Sub-hydrostatic pore pressure.
Barail Group	3,060.0 m	3,450.0 m	Interbedded marine shales, carbonaceous coals & gas sands.

SECTION 8: HISTORICAL DRILLING HAZARDS & NPT LOG (PAGE 147)

INCIDENT 1: DIFFERENTIAL PIPE STICKING AT 1,680m MD (GIRUJAN CLAY)

While pulling out of hole (POOH) for directional survey at 1,680m MD, drill string was kept stationary for 3 hours 15 minutes due to top-drive hydraulic hose leak. Upon resuming hoisting, string was found differentially stuck with overpull exceeding 110,000 lbs. Mud weight was 10.9 ppg water-based polymer mud with excessive filter cake thickness (8/32"). Jarred string with maximum safe pull for 4 hours without release.

FIELD-PROVEN MITIGATION & SOP: Mixed and spotted 50 bbl Oil-Based Mud (OBM) lubricant soak pill weighted to 11.0 ppg with 4% pipe-release surfactant. Allowed soak time of 12 hours. String jarred free at 08:30 hrs. Total NPT: 14 days (INR 8.0 Crore). Recommendation: Maintain continuous string rotation (>60 RPM) in Girujan. Do not allow stationary pipe to exceed 5 minutes. Use KCl-polymer mud system with 5% KCl for clay inhibition.

INCIDENT 2: LOST CIRCULATION IN TIPAM SANDSTONE AT 1,840m MD

Encountered severe loss of circulation at 1,840m MD in Tipam Sandstone TS-3 pay interval. Returns dropped to 35% with 60 bbl/hr loss rate. Pumped 25 bbl coarse calcium carbonate (CaCO3) + mica LCM pill. Circulation fully restored after 4 hours soak. Maintained MW below 10.8 ppg.

STATUTORY SIGN-OFF: Verified by P. Saikia (Chief Drilling Specialist, Oil India Ltd.) · OISD-STD-174 Compliant